

# Axion Electrodynamics in FDTD

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FDTD is used to solve the axion-modified Maxwell's equations numerically. The results in 1D are stable and align with theoretical expectation where axion field simply propagates as a free wave; while there are complications in 2D due to difficulties in implementing total field scattered field (TFSF) in 2D.

## I. INTRODUCTION

The axion was originally introduced by Roberto Peccei and Helen Quinn [1] to resolve the longstanding strong CP problem. CP violation is predicted by SM and one of the necessary conditions for matter antimatter asymmetry in the early universe [2], therefore, it is expected to be observed. It was indeed observed indirectly in 1964 by [3] with the Fitch-Cronin experiment, where they found that CP symmetry was broken via weak interactions from kaon decay. But strangely, no CP violation has ever been observed in strong interactions.

The strong CP problem is often restated as the nondetection of neutron electric dipole moment (nEDM), as this would imply CP symmetry is preserved in the QCD sector. More specifically, the term in the QCD Lagrangian that predicts CP violation is as follows:

$$\mathcal{L}_{QCD} \supset \Theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}_a^{\mu\nu} \quad (1)$$

where  $g_s$  is the QCD coupling constant,  $G$  and  $\tilde{G}$  are the gluon field strength and its dual respectively.  $\Theta$  is a constant parameter that gives rise to an EDM in neutron and the nondetection of EDM in neutron implies,  $|\Theta| < 0.8 \times 10^{-10}$  [4]. This arbitrarily small  $\Theta$  is a fine tuning problem.

One possible resolution to the strong CP problem was proposed by Peccei and Quinn. They postulated a global chiral symmetry  $U(1)_{PQ}$  and a new scalar field  $\theta$ . When this symmetry is spontaneously broken at some energy scale  $f_a$ ,  $\theta$  settles in a new minimum and obtains a VEV. The corresponding pseudo Nambu-Goldstone boson is a massless axion field and replaces the problematic term in the QCD Lagrangian:

$$\mathcal{L}_a \supset \frac{N_{DW}\theta}{f_a} \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}_a^{\mu\nu} \quad (2)$$

The only change from Eq. (1) to Eq. (2) is that now the constant  $\Theta$  becomes a value the axion field  $\theta(\mathbf{x}, t)$  takes, scaled by its decay rate  $f_a$  and a constant  $N_{DW}$ .

One of the consequences of this construction is the introduction of an interaction vertex  $a - \gamma - \gamma$  as shown in Fig. 1. This allows photon and axion to oscillate into one another, opening a small window of possibility of detecting axions. This requires a good understanding of the behaviours of axions. In this report, I present the numerical solution to the axion-modified Maxwell's equations.

## II. MODIFIED MAXWELL'S EQUATIONS

The axion photon coupling term in the Lagrangian can be written as

$$\mathcal{L}_\theta = -\frac{\kappa}{\mu_0 c} \theta \mathbf{E} \cdot \mathbf{B} = \frac{\kappa}{4\mu_0} \theta F_{\mu\nu} \tilde{F}^{\mu\nu} \quad (3)$$

where  $\mu_0$  is the vacuum permeability,  $c$  is the speed of light,  $F$  and  $\tilde{F}$  are the electromagnetic tensor and its dual. An additional term in the Lagrangian also means a different sets of equation of motions (EOM) [6]

$$\begin{aligned} \partial_\mu F^{\mu\nu} - \kappa \partial_\mu (\theta \tilde{F}^{\mu\nu}) &= \mu_0 J_e^\nu \\ \square \theta &= \frac{\kappa}{4\mu_0} F_{\mu\nu} \tilde{F}^{\mu\nu} - \frac{\partial U(\theta)}{\partial \theta} \end{aligned} \quad (4)$$

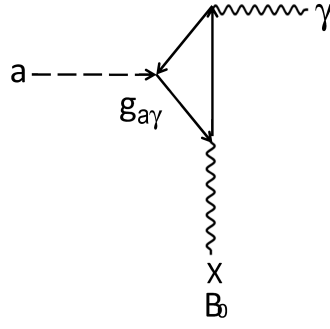


FIG. 1. The interaction vertex for axion-photon coupling under an external B field.[5]

where  $J_e^\nu$  is the electric 4-current  $(\rho, \mathbf{J})$ ,  $U(\theta)$  is the axion potential. There were efforts deriving the modified Maxwell's equations [6][7] [8], but they adopted different conventions and assumptions. The EOMs in Eq. (4) are obtained by using the Euler-Lagrange equations, but it does not give the EOM for the dual  $\tilde{F}^{\mu\nu}$ . In the original Maxwell's equations, the EOM for the dual is conventionally taken to be

$$\partial_\mu \tilde{F}^{\mu\nu} = 0 \quad (5)$$

because of the non-observation of magnetic monopole; but this makes Eq. (4) no longer invariant under dual transformation

$$\begin{pmatrix} \mathbf{E}' \\ \mathbf{B}' \end{pmatrix} = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{B} \end{pmatrix} \quad (6)$$

In principle, the electromagnetic (EM) field does not have to be dual invariant in the presence of an axion field, but an axion theory with such an invariance would be much cleaner. Therefore, I follow the assumptions used by Visinelli [6] such that Eq. (5) is no longer true, thus obtaining a new set of EOMs

$$\begin{aligned} \partial_\mu F^{\mu\nu} - \kappa \partial_\mu (\theta \tilde{F}^{\mu\nu}) &= \mu_0 J_e^\nu \\ \partial_\mu \tilde{F}^{\mu\nu} + \kappa \partial_\mu (\theta F^{\mu\nu}) &= \mu_0 J_m^\nu / c \\ \square \theta &= \frac{\kappa}{4\mu_0} F_{\mu\nu} \tilde{F}^{\mu\nu} - \frac{\partial U(\theta)}{\partial \theta} \end{aligned} \quad (7)$$

with  $J_m^\nu$  being the magnetic counterpart of  $J_e^\nu$ .

Rewriting Eq. (7) in  $\mathbf{E}$  and  $\mathbf{B}$  in natural units,

$$\begin{aligned} \nabla \cdot (\mathbf{E} - \kappa \theta \mathbf{B}) &= 0 \\ \nabla \cdot (\mathbf{B} + \kappa \theta \mathbf{E}) &= 0 \\ \nabla \times (\mathbf{E} + \kappa \theta \mathbf{B}) &= -\frac{\partial}{\partial t} (\mathbf{B} + \kappa \theta \mathbf{E}) \\ \nabla \times (\mathbf{B} + \kappa \theta \mathbf{E}) &= \frac{\partial}{\partial t} (\mathbf{E} + \kappa \theta \mathbf{B}) \\ (\square + m^2) \theta &= -\kappa \mathbf{E} \cdot \mathbf{B} \end{aligned} \quad (8)$$

The potential  $U(\theta)$  is not fixed by the theory, one example is in [9], but in most cases, the dominant term can be approximated to be  $\frac{1}{2} m^2 \theta^2$  [10], whose derivative is evaluated to give the  $m^2$  in the last line of Eq. (8).

### III. FDTD ALGORITHM

#### A. One-dimensional FDTD

The modified Maxwell's equations are really just the original Maxwell's equations but in terms of  $\hat{\mathbf{E}}$  and  $\hat{\mathbf{B}}$  where

$$\begin{aligned}\hat{\mathbf{E}} &= (\mathbf{E} - \kappa\theta\mathbf{B}) \\ \hat{\mathbf{B}} &= (\mathbf{B} + \kappa\theta\mathbf{E}).\end{aligned}\tag{9}$$

The reverse relations are,

$$\begin{aligned}\mathbf{E} &= \frac{\hat{\mathbf{E}} + \kappa\theta\hat{\mathbf{B}}}{1 + \kappa^2\theta^2} \\ \mathbf{B} &= \frac{\hat{\mathbf{B}} - \kappa\theta\hat{\mathbf{E}}}{1 + \kappa^2\theta^2}\end{aligned}\tag{10}$$

In the rest of this report, the hats in all the component equations will be dropped, the physical fields will be written as  $E_{phy,i}$  in  $i$  direction. Explicitly,  $\hat{\mathbf{E}} = (E_x, E_y, E_z)$  and  $\mathbf{E} = (E_{phy,x}, E_{phy,y}, E_{phy,z})$ . For concreteness, the initial  $\mathbf{E}$  and  $\mathbf{B}$  are set to propagate only in the  $z$  direction and only has a  $x$  component and  $y$  component respectively. However, due to the dependence of  $\hat{\mathbf{E}}$  on both  $\mathbf{E}$  and  $\mathbf{B}$ ,  $\hat{\mathbf{E}}$  and  $\hat{\mathbf{B}}$  will in general have two components as they propagate. The modified Maxwell's equations in 1D are reduced to,

$$\begin{aligned}\frac{\partial E_x}{\partial t} &= -\frac{\partial B_y}{\partial z} \\ \frac{\partial E_y}{\partial t} &= \frac{\partial B_x}{\partial z} \\ \frac{\partial B_x}{\partial t} &= \frac{\partial E_y}{\partial z} \\ \frac{\partial B_y}{\partial t} &= -\frac{\partial E_x}{\partial z} \\ \frac{\partial^2 \theta}{\partial t^2} &= \frac{\partial^2 \theta}{\partial z^2} - \kappa \left[ \left( \frac{E_x + \kappa\theta B_x}{1 + \kappa^2\theta^2} \right) \left( \frac{B_x - \kappa\theta E_x}{1 + \kappa^2\theta^2} \right) - \left( \frac{E_y + \kappa\theta B_y}{1 + \kappa^2\theta^2} \right) \left( \frac{B_y - \kappa\theta E_y}{1 + \kappa^2\theta^2} \right) \right] - m^2\theta.\end{aligned}\tag{11}$$

where the dot product in the last line of Eq. (8) is expanded in written in terms of the hat fields as in Eq. (10). In FDTD, derivatives are approximated as central differences, so Eq. (11) becomes,

$$\begin{aligned}E_x^{n+1/2}[k] &= E_x^{n-1/2}[k] + \frac{\Delta t}{\Delta x} (B_y^n[k-1/2] - B_y^n[k+1/2]) \\ E_y^{n+1/2}[k] &= E_y^{n-1/2}[k] + \frac{\Delta t}{\Delta x} (B_x^n[k+1/2] - B_x^n[k-1/2]) \\ B_x^{n+1}[k+1/2] &= B_x^n[k+1/2] + \frac{\Delta t}{\Delta x} (E_y^{n+1/2}[k+1] - E_y^{n+1/2}[k]) \\ B_y^{n+1}[k+1/2] &= B_y^n[k+1/2] + \frac{\Delta t}{\Delta x} (E_x^{n+1/2}[k] - E_x^{n+1/2}[k+1]) \\ \theta^{n+1}[k] &= 2\theta^n[k] - \theta^{n-1}[k] + \left( \frac{\Delta t}{\Delta x} \right)^2 (\theta^n[k+1] - 2\theta^n[k] + \theta^n[k-1]) \\ &\quad - (\Delta t)^2 \left( E_{phy,x}^{n+1/2}[k] B_{phy,x}^{n+1}[k+1/2] + E_{phy,y}^{n+1/2}[k] B_{phy,y}^{n+1}[k+1/2] \right) - (\Delta t)^2 m^2 \theta^n[k].\end{aligned}\tag{12}$$

#### B. Two-dimensional FDTD

By the same token, the FDTD algorithm can be extended to 2D. Again, because of Eq. (10), all three components of  $\hat{\mathbf{E}}$  are generally non-zero in 2D. The source is injected in  $E_z$  and  $\hat{\mathbf{E}}$  is set to propagate on the  $\hat{x} - \hat{y}$  plane. The 2D Maxwell's equations are

$$\begin{aligned}
\frac{\partial E_x}{\partial t} &= \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \\
\frac{\partial E_y}{\partial t} &= \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \\
\frac{\partial E_z}{\partial t} &= \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \\
\frac{\partial B_x}{\partial t} &= \frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \\
\frac{\partial B_y}{\partial t} &= \frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \\
\frac{\partial B_z}{\partial t} &= \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \\
\frac{\partial^2 \theta}{\partial t^2} &= \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} \\
&- \kappa (E_{phy,x} B_{phy,x} + E_{phy,y} B_{phy,y} + E_{phy,z} B_{phy,z}) - m^2 \theta.
\end{aligned} \tag{13}$$

and the update equations are now,

$$\begin{aligned}
E_x^{n+1/2}[i, j] &= E_x^{n-1/2}[i, j] + \frac{\Delta t}{\Delta x} (B_z^n[i, j+1/2] - B_z^n[i, j-1/2]) \\
E_y^{n+1/2}[i, j] &= E_y^{n-1/2}[i, j] + \frac{\Delta t}{\Delta x} (B_z^n[i-1/2, j] - B_z^n[i+1/2, j]) \\
E_z^{n+1/2}[i, j] &= E_z^{n-1/2}[i, j] + \frac{\Delta t}{\Delta x} (B_y^n[i+1/2, j] - B_y^n[i-1/2, j] + B_x^n[i, j-1/2] - B_x^n[i, j+1/2]) \\
B_x^{n+1}[i, j+1/2] &= B_x^n[i, j+1/2] + \frac{\Delta t}{\Delta x} (E_z^{n+1/2}[i, j] - E_z^{n+1/2}[i, j+1]) \\
B_y^{n+1}[i+1/2, j] &= B_y^n[i+1/2, j] + \frac{\Delta t}{\Delta x} (E_z^{n+1/2}[i+1, j] - E_z^{n+1/2}[i, j]) \\
B_z^{n+1}[i+1/2, j+1/2] &= B_z^n[i+1/2, j+1/2] + \frac{\Delta t}{\Delta x} (E_x^{n+1/2}[i, j+1] - E_x^{n+1/2}[i, j] + E_y^{n+1/2}[i, j] - E_y^{n+1/2}[i+1, j]) \\
\theta^{n+1}[i, j] &= 2\theta^n[i, j] - \theta^{n-1}[i, j] \\
&+ \left(\frac{\Delta t}{\Delta x}\right)^2 (\theta^n[i+1, j] + \theta^n[i, j+1] - 4\theta^n[i, j] + \theta^n[i-1, j] + \theta^n[i, j-1]) \\
&- (\Delta t)^2 \kappa (E_{phy,x}^{n+1/2}[i, j] B_{phy,x}^{n+1}[i, j+1/2] + E_{phy,y}^{n+1/2}[i, j] B_{phy,y}^{n+1}[i+1/2, j] + E_{phy,z}^{n+1/2}[i, j] B_{phy,z}^{n+1}[i+1/2, j+1/2]) \\
&- (\Delta t)^2 m^2 \theta^n[k].
\end{aligned} \tag{14}$$

For stability in both 1D and 2D,  $\Delta t/\Delta x$  is taken to be 0.5 so that  $(\Delta t/\Delta x)^2 < 0.5 = 0.25$ , satisfying the Courant-Friedrichs-Lewy (CFL) stability conditions. For a plane wave source, the number of points sampled in one wavelength is specified and will be discussed in Sec. IV in detail, which imposes a constraint on the range of  $\Delta t$  that can be taken. As stated in Eq. (3), axion conversion is possible only if there is a non-zero dot product. Therefore, only a  $\mathbf{E}$  source is not sufficient, and external  $\mathbf{B}$  parallel to the incident  $\mathbf{E}$  is necessary for the conversion to happen (also see Fig. 1). In both cases, a constant  $\mathbf{B}_{\text{external}}$  is injected parallel to the incident  $\mathbf{E}$ .

## IV. RESULTS

### A. Femto-second pulse (1D)

In this first example, a femto-second pulse source was studied so that a direct comparison to the results from assignment 5 can be made. Since  $E = B/c$ ,  $|B_{\text{external}}|$  is usually extremely small compared to  $|E_{\text{source}}|$ . For demonstration,  $|B_{\text{external}}|$  is taken to have a similar order of magnitude as  $|E_{\text{source}}|$ .

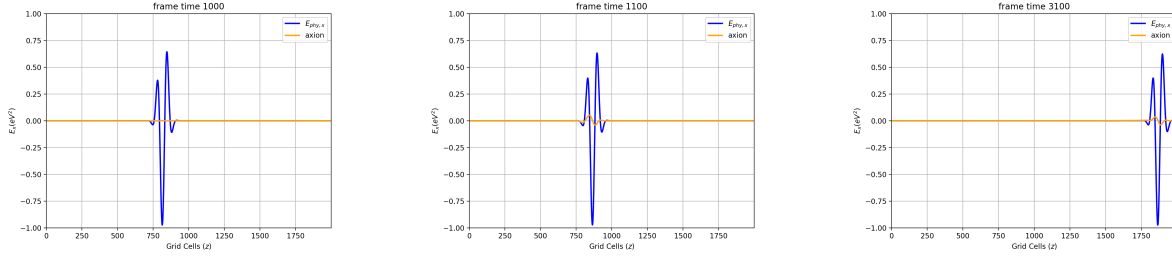


FIG. 2. Axion conversion with a pulse source in 1D. The external B field is turned on at  $t = 1000$ . Axion conversion occurs right after the external B field was turned on (middle panel) and it propagates steadily throughout.

### B. Plane waves (1D)

One of the popular axion detection experiments is axion detection from the Sun. Fig. 3 shows a common experimental set-up in detecting axion by converting photons from the sun with strong B field. Motivated from such experiments, more realistic  $\mathbf{E}_{\text{source}}$  and  $\mathbf{B}_{\text{external}}$  values are used. The core temperature of the sun is roughly 30 million K, which corresponds to photon energies  $\sim 3\text{keV}$ . With an average solar flux  $S = 1.4\text{kW/m}^2$ ,  $|E_{\text{source}}| = \sqrt{\mu_0 c S_{\text{avg}}} \approx 700\text{V/m}$ .  $\mathbf{B}_{\text{external}}$  is set to 9T, which is the strength used in typical axion experiments [11]. The absolute intensity of  $\mathbf{E}_{\text{source}}$  is unimportant but the ratio of  $|E_{\text{source}}|/|B_{\text{external}}|$ , such that a realistic conversion amplitude can be simulated.

As shown in Fig. 4, the axion amplitude is much lower, with  $\frac{|\theta|}{|E_{\text{source}}|} \approx 3.33 \times 10^{-13}$ , reflecting the rarity of axion conversion, hence the difficulty of axion detection.

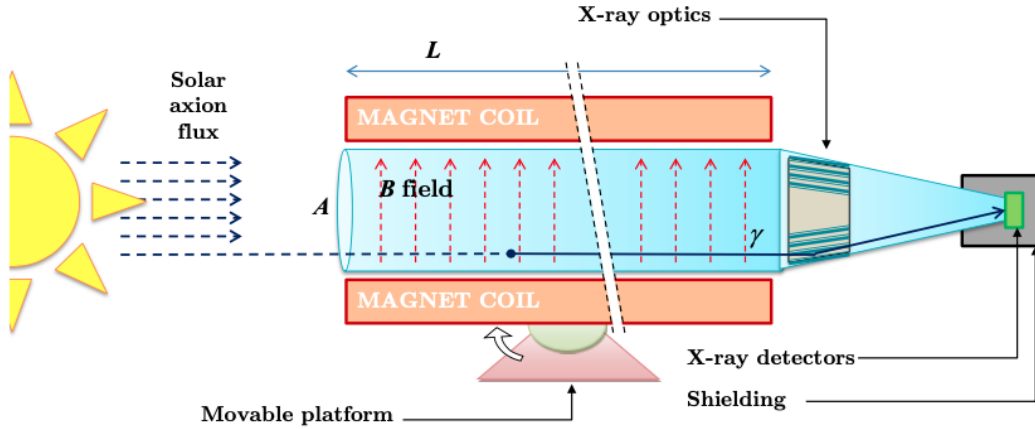


FIG. 3. A common experimental set-up for a helioscope. Figure from [12]

### C. FDTD in 2D

FDTD in 2D faces complication at the point of source injection, which leads to serious instability problem. For a circular wave source, even without axion,  $\mathbf{B}$  diverges at the point of source injection (see left panel of Fig. 6). The resultant  $E_z$  is stable (right panel of Fig. 6), this is because the instability occurs in x and y components, which do not feedback to the update of  $E_z$ . However with axion, the divergence in B renders  $E_z$  unstable due to Eq. (9).

Without axion, the instability can be avoided by using plane waves in 2D as shown in Fig. 6. With axion,  $E_z$  goes up to  $O(10)$ , it can be seen on the right panel of Fig. 7 that the colorbar goes up to 20, despite not visually obvious from the density plot. One possible explanation might be the boundary effect. As this simulation is running only for

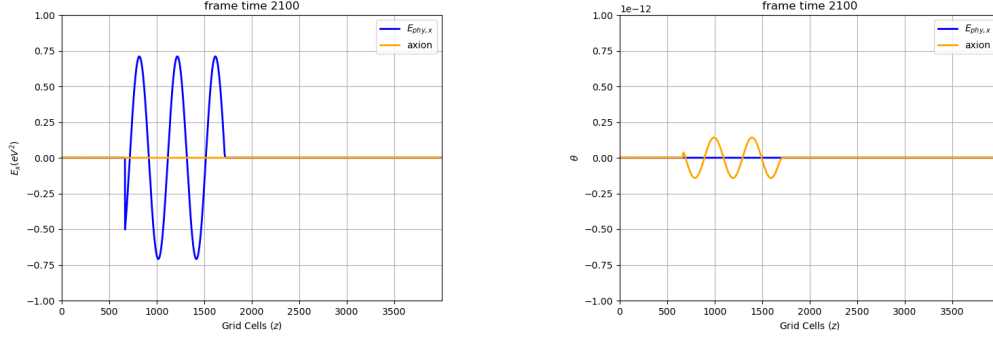


FIG. 4. Axion conversion with plane wave source in 1D. The external B field is turned on at  $t = 2000$ . The amplitude of axion wave (left panel) is much lower  $10^{-12}$ .

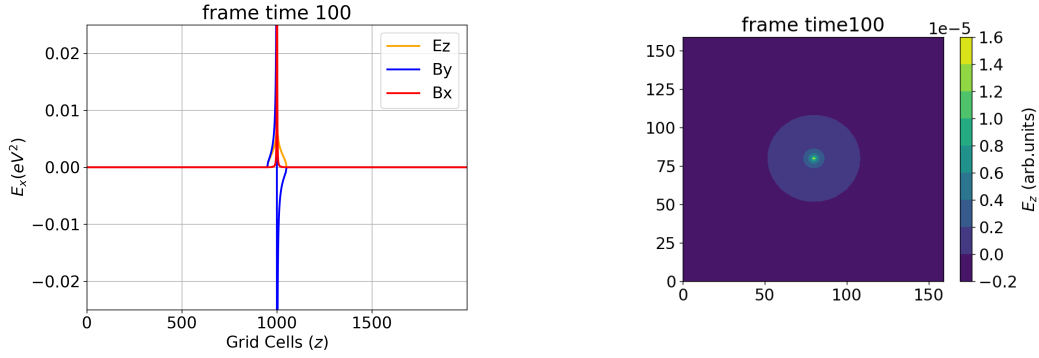


FIG. 5. Circular wave source. Left: Both Bx and By diverge at the point of injection (a horizontal slice in the middle of the domain). Right: The result E is stable.

1000 timesteps, where the field touches the boundary at the end of the simulation, the problem could be verified by running for longer time. However, absorbing boundary conditions in 2D has to be implemented.

## V. CONCLUSIONS

In the 1D case, as expected theoretically, photons could be converted into axions under a magnitude that is parallel to the E field as indicated in the dot product in the last line of Eq. (8). Without such a parallel component, axion cannot be produced, this can be verified by turning  $\mathbf{B}_{\text{external}}$  off or injecting  $\mathbf{B}_{\text{external}}$  in the direction perpendicular to  $\mathbf{E}_{\text{source}}$ . However, instabilities occur in 2D due to the divergence of  $\mathbf{B}$  at the point of  $\mathbf{E}_{\text{source}}$  injection. This could possibly be avoided by implementing TFSF in 2D. For example,  $\mathbf{B}_{\text{external}}$  can be applied on outside certain radius, and  $\mathbf{E}_{\text{source}}$  could be injected outside that radius. However due to time limitation, this was not done.

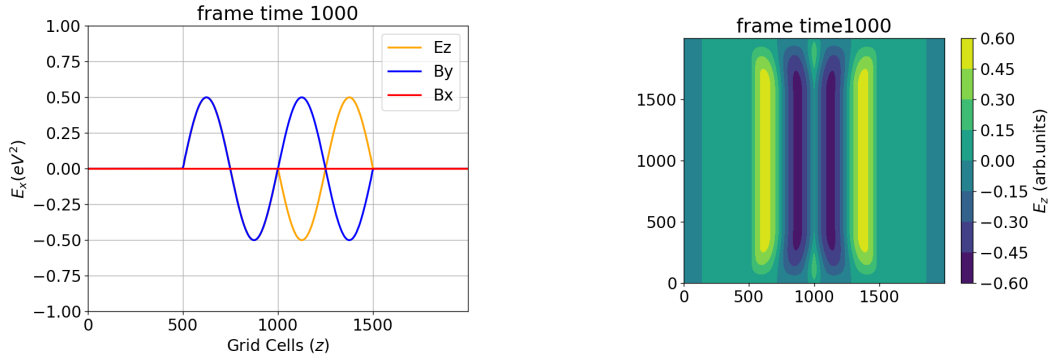


FIG. 6. Plane wave source. Left: Bx and By did not diverge. (a horizontal slice in the middle of the domain) Right: The result E is also stable.

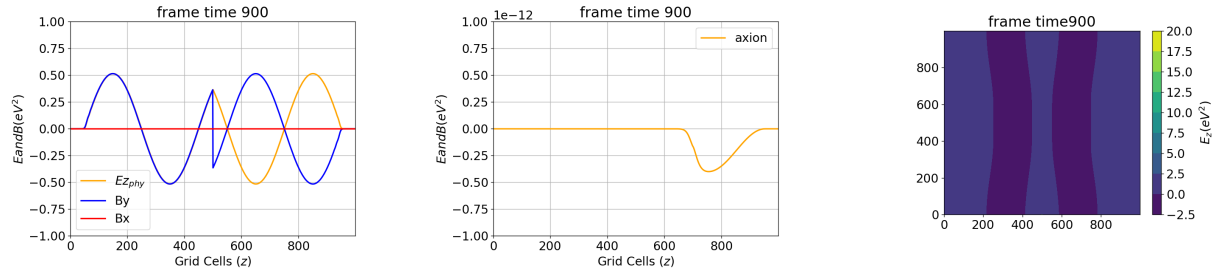


FIG. 7. Axion conversion with plane waves in 2D. The external B field is turned on at  $t = 300$ . The left and middle panel show a horizontal slice in the middle of the domain. Left: E and B field. Middle: Axion field. Right:  $E_{phy,z}$

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